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Support device and foldable device

Abstract

A support device is provided. The support device is disposed on a back side of a bendable structure, the bendable structure comprising a bending portion, and a first non-bending portion and a second non-bending portion disposed on two sides of the bending portion, respectively; the support device comprising: a first support, configured to be connected to the first non-bending portion; a second support, configured to be connected to the second non-bending portion; a rotating shaft support, configured to be rotatably connected to the first support and the second support; and at least one support component, wherein each support component comprises a strip-shaped support bar, wherein the support bar is connected to the rotating shaft support and configured to be fixedly connected to the bending portion.

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Background/Summary

(1) This application is a continuation of U.S. patent application Ser. No. 17/781,504, filed on Jun. 1, 2022, which is a 371 of PCT Application No. PCT/CN2021/104534, filed on Jul. 5, 2021, and claims priority to Chinese Patent Application No. 202010824603.9, filed on Aug. 17, 2020 and entitled “SUPPORT DEVICE AND FOLDABLE DEVICE”, the content of which is herein incorporated by reference in its entirety.

TECHNICAL FIELD
(1) The present disclosure relates to the field of machinery technologies, and in particular relates to a support device and a foldable device.
BACKGROUND

(2) With the development of science and technology, flexible display substrates are widely applied in foldable display devices due to their good bending performances.

SUMMARY

(3) Embodiments of the present disclosure provide a support device and a foldable device. The technical solutions are as follows.

(4) In an aspect, a support device is provided. The support device is disposed on a back side of a bendable structure, the bendable structure includes a bending portion, and a first non-bending portion and a second non-bending portion disposed on two sides of the bending portion, respectively. The support device includes: a first support, configured to be connected to the first non-bending portion; a second support, configured to be connected to the second non-bending portion; a rotating shaft support, configured to be rotatably connected to the first support and the second support; and at least one support component, wherein each support component is connected to the rotating shaft support and configured to be connected to the bending portion.

(5) Optionally, the rotating shaft support includes a strip-shaped support beam and a strip-shaped rotating shaft; wherein the support beam is provided with a rotating shaft hole penetrating through the support beam, wherein an axis of the rotating shaft hole is parallel to an extending direction of the support beam; and the rotating shaft is disposed in the rotating shaft hole and is rotatable in the rotating shaft hole.

(6) Optionally, each support component includes a strip-shaped support bar; wherein one end of the support bar is bonded to the support beam, and another end of the support bar is configured to be fixedly connected to the bending portion.

(7) Optionally, the support beam is provided with at least one first groove, wherein one end of the support bar is disposed in one first groove and is bonded to the first groove.

(8) Optionally, an orthographic projection of one end of the support bar on the support beam is within an orthographic projection of another end of the support bar on the support beam.

(9) Optionally, the support bar has a T-shaped cross section, and the cross section is perpendicular to an extending direction of the support bar.

(10) Optionally, each support component includes a strip-shaped support bar and a plurality of connectors, each of the connectors being bent-shaped; wherein one end of each of the connectors is fixedly connected to the support bar, and another end of the connector is connected to the support beam, and at least two connectors of the plurality of connectors are disposed on different sides of the support bar; and the support bar is further configured to be fixedly connected to the bending portion.

(11) Optionally, the support beam is provided with a plurality of second grooves, and each support component further includes a first protruding structure connected to another end of the connector, the first protruding structure being disposed in one of the second grooves.

(12) Optionally, each support component includes a strip-shaped support bar, a sheet structure connected to a side of the support bar, and a second protruding structure at an end, distal from the support bar, of the sheet structure; and the support beam is provided with a third groove, a fourth groove, and a communication groove disposed inside the support beam and configured to communicate the third groove and the fourth groove; wherein the support bar is disposed in the third groove, the second protruding structure is disposed in the fourth groove, and the sheet structure is disposed in the communication groove; and wherein the support bar is further configured to be fixedly connected to the bending portion.

(13) Optionally, a length of the third groove along the extending direction of the support beam is greater than a total length of the support bar and the sheet structure along an extending direction of the support bar.

(14) Optionally, a bottom surface of the third groove and a bottom surface of the fourth groove are both arc surfaces, and a surface, proximal to the support beam, of the support bar and a surface, proximal to the support beam, of the sheet structure are both the arc surfaces; wherein the arc

surfaces are protruded in a direction proximal to the bendable structure, and a radius of the arc surfaces is greater than a radius of the rotating shaft hole.
(15) Optionally, the rotating shaft support further includes: a strip-shaped support structure connected to the support beam and two plate-shaped connecting structures, the two connecting structures being connected to two ends of the support structure, respectively; wherein each of the connecting structures is at least provided with a first through hole, and a second through hole and a third through hole disposed on two sides of the first through hole respectively, wherein an axis of the first through hole is co-linear with an axis of the rotating shaft hole, the first through hole is connected to the rotating shaft, the second through hole is connected to the first support, and the third through hole is connected to the second support.
(16) Optionally, the support device further includes: a first bracket and a second bracket; wherein one end of the first bracket is rotatably connected to the rotating shaft support, and another end of the first bracket is fixedly connected to the first support; and one end of the second bracket is rotatably connected to the rotating shaft support, and another end of the second bracket is fixedly connected to the second support.
(17) In another aspect, a foldable device is provided. The foldable device includes: a bendable structure, and at least one support device described in the above aspect; wherein the support device is disposed on and connected to a back side of the bendable structure.
(18) Optionally, the bendable structure is a flexible display panel.
(19) Optionally, the foldable device further includes: a reinforcement plate; wherein the support device and the bendable structure are connected by the reinforcement plate.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) For clearer descriptions of the technical solutions in the embodiments of the present disclosure, the following briefly introduces the accompanying drawings required for describing the embodiments. Apparently, the accompanying drawings in the following description show merely some embodiments of the present disclosure, and persons of ordinary skill in the art may still derive other drawings from these accompanying drawings without creative efforts.
(2) FIG. **1** is a schematic diagram of a foldable display device being outwardly folded in the related art;
(3) FIG. **2** is a schematic diagram of a foldable display device being inwardly folded in the related art;
(4) FIG. **3** is a schematic structural diagram of a support device according to embodiments of the present disclosure;
(5) FIG. **4** is a schematic structural diagram of a bendable structure according to embodiments of the present disclosure;
(6) FIG. **5** is a schematic structural diagram of a rotating shaft support according to embodiments of the present disclosure;
(7) FIG. **6** is a schematic structural diagram of a support beam and a support component according to embodiments of the present disclosure;
(8) FIG. **7** is a schematic structural diagram of a support component according to embodiments of the present disclosure;
(9) FIG. **8** is a schematic structural diagram of another rotating shaft support according to embodiments of the present disclosure;
(10) FIG. **9** is a schematic structural diagram of another support beam and support component according to embodiments of the present disclosure;
(11) FIG. **10** is a schematic structural diagram of another support component according to

embodiments of the present disclosure;
(12) FIG. **11** is a schematic structural diagram of still another rotating shaft support according to embodiments of the present disclosure;
(13) FIG. **12** is a schematic diagram of a partial structure of the rotating shaft support shown in FIG. **11**;
(14) FIG. **13** is a schematic structural diagram of still another support beam and support component according to embodiments of the present disclosure;
(15) FIG. **14** is a schematic structural diagram of still another support component according to embodiments of the present disclosure;
(16) FIG. **15** is a schematic structural diagram of yet another rotating shaft support according to embodiments of the present disclosure;
(17) FIG. **16** is a schematic diagram of a partial structure of the support beam in the rotating shaft support shown in FIG. **15**;
(18) FIG. **17** is a sectional view of the support beam in the rotating shaft support shown in FIG. **15** along an AA direction;
(19) FIG. **18** is a sectional view of the support beam and support component shown in FIG. **13** along a BB direction;
(20) FIG. **19** is a schematic structural diagram of another support device according to embodiments of the present disclosure;
(21) FIG. **20** is a schematic structural diagram of a foldable device according to embodiments of the present disclosure; and
(22) FIG. **21** is an exploded schematic diagram of the foldable device shown in FIG. **20**.

DETAILED DESCRIPTION

(23) In order to make the object, technical solutions and advantages of the present disclosure clearer, the embodiments of the present disclosure are described in detail below with reference to the accompanying drawings.
(24) In the related art, a foldable display device includes a flexible display panel and a support device disposed on a side of the flexible display panel. The flexible display panel has a bending portion and non-bending portions disposed on two sides of the bending portion respectively. The support device may include two support parts, and each support part may be connected to one non-bending portion of the flexible display panel to support the non-bending portion.
(25) However, the support device does not support the bending portion of the flexible display panel, and the flexible display panel has a poor rigidity. Thus, the bending portion of the flexible display panel bulges easily, resulting in a poor flatness of the flexible display panel.
(26) In the field of display technologies, the foldable display device may be prepared by using a bendable flexible display panel, so as to meet the users' demand for large screen display and improve portability of the display device. The foldable display device is foldable and occupies a relatively small space after being folded, and thus is portable. When the foldable display device is unfolded, the region for displaying images has a relatively large area, and the display effect is good.
(27) Generally, the flexible display panel in the foldable display device needs to be supported such that the flexible display panel has a good flatness and impact resistance, thereby ensuring the display effect of the foldable display device. However, in the related art, the support device cannot provide the bending portion of the flexible display panel effectively, resulting in poor rigidity of the bending portion of the flexible display panel. Therefore, in the case that the foldable display device is bent, the bending portion of the flexible display panel bulges backwards easily, which results in poor flatness of the flexible display panel when the foldable display device returns from the bent state to the flat state. Moreover, as the bending portion of the flexible display panel bulges backwards, the creases between the bending portion and the non-bending portion are serious, and the film layer in the flexible display panel is easy to strip off. Therefore, the flexible display panel as a short service life.

(28) Exemplarily, referring to FIG. **1**, in the case that the foldable display device **00** is folded outwardly, the bending portion **001***a* of the flexible display panel **001** bulges towards the side proximal to the support device **002** supporting the flexible display panel **001**. Alternatively, referring to FIG. **2**, in the case that the foldable display device **00** is folded inwardly, the bending portion **001***a* of the flexible display panel **001** bulges towards the side distal from the support device **002** supporting the flexible display panel **001**.
(29) After the foldable display device **00** is folded outwardly, the flexible display panel **001** is disposed at the outer side relative to the support device **002**. That is, the display surface of the flexible display panel **001** is disposed at the outer side. After the foldable display device is folded inwardly, the flexible display panel **001** is disposed at the inner side relative to the support device **002**. That is, the display surface of the flexible display panel **001** is disposed at the inner side.
(30) FIG. **3** is a schematic structural diagram of a support device according to embodiments of the present disclosure. The support device **101** may be arranged on the back side of the bendable structure (not shown in FIG. **3**) to support the bendable structure. Here, the bendable structure may be a flexible display panel. Referring to FIG. **4**, the bendable structure **102** may include a bending portion **1021**, and a first non-bending portion **1022** and a second non-bending portion **1023** which are disposed on two sides of the bending portion **1021**, respectively.
(31) Referring to FIG. **3**, the support device **101** may include a first support **1011**, a second support **1012**, a rotating shaft support **1013**, and at least one support component **1014**. Exemplarily, FIG. **3** shows three support components **1014**.
(32) The first support **1011** may be configured to be connected to the first non-bending portion **1022** of the bendable structure **102** for supporting the first non-bending portion **1022**. The second support **1012** may be configured to be connected to the second non-bending portion **1023** of the bendable structure **102** for supporting the second non-bending portion **1023**. The rotating shaft support **1013** may be disposed between the first support **1011** and the second support **1012**, and is rotatably connected to the first support **1011** and the second support **1012**. Each support component **1014** may be connected to the rotating shaft support **1013**, and connected to the bending portion **1021**. Each support component **1014** may be configured to support the bending portion **1021** of the bendable structure **102**.
(33) In the case that the bendable structure **102** is supported by the support device **101** according to the embodiments of the present disclosure, because the rotating shaft support **1013** is rotatably connected to the first support **1011** and the second support **1012**, the first support **1011** can drive the first non-bending portion **1022** to rotate relative to the bending portion **1021**, and the second support **1012** can drive the second non-bending portion **1023** to rotate relative to the bending portion **1021**, thereby folding the foldable device.
(34) Moreover, in the case that the foldable device is folded, since the support component **1014** in the support device **101** can support the bending portion **1021** of the bendable structure **102** so as to provide support for the bending portion **1021**, the bending portion **1021** can be prevented from bulging, thereby ensuring the flatness of the bendable structure **102**.
(35) In the embodiments of the present disclosure, the bendable structure **102** may be a flexible display panel. The support device **101** is disposed on the back side of the flexible display panel such that the support device **101** can support the flexible display panel **102**. Moreover, in the case that the bendable structure **102** is a flexible display panel, the foldable device may be a foldable display device. Since the support device **101** is capable of supporting all regions of the flexible display panel **102**, the bending portion **1021** of the flexible display panel **102** can be prevented from bulging when the foldable display device consisting of the flexible display panel **102** and the support device **101** is folded. In addition, the flexible display panel **102** has a good flatness when the foldable display device returns from the bent state to the flat state. Certainly, the bendable structure **102** may also be other bendable structures. For example, the bendable structure **102** may be a flexible plate structure, which is not limited in the embodiments of the present disclosure.

(36) In summary, the embodiments of the present application provide a support device. The rotating shaft support and each support component in the support device can provide support for the bending portion of the bendable structure. Moreover, the first support in the support device can provide support for the first non-bending portion of the bendable structure, and the second support in the support device can provide support for the second non-bending portion of the bendable structure. That is, the support device according to the embodiments of the present disclosure can be configured to support the entire bendable structure, which can prevent the bending portion of the bendable structure from bulging, thereby ensuring the flatness of the bendable structure.
(37) FIG. **5** is a schematic structural diagram of a rotating shaft support according to the embodiments of the present disclosure. Referring to FIG. **5**, the rotating shaft support **1013** may include a strip-shaped support beam **10131**, and a strip-shaped rotating shaft. The support beam **10131** may have a rotating shaft hole **10131***a* penetrating through the support beam **10131**, and an axis of the rotating shaft hole **10131***a* may be parallel to an extending direction X of the support beam **10131**. The rotating shaft may be disposed in the rotating shaft hole **10131***a*, and is capable of rotating in the rotating shaft hole **10131***a*, such that the rotating shaft support **1013** rotates relative to the first support **1011** and the second support **1012**. In order to clearly illustrate the rotating shaft hole **10131***a* in the support beam **10131**, the rotating shaft is not shown in FIG. **5**.
(38) Optionally, referring to FIG. **5**, the cross section of the support beam **10131** may be rectangular, and the cross section may be perpendicular to the extending direction X of the support beam **10131**. Certainly, the cross section of the support beam **10131** may also be of other shapes, for example, may be circular, which is not limited in the embodiments of the present disclosure.
(39) As an optional implementation, FIG. **6** is a schematic structural diagram of a support beam and a support component according to the embodiments of the present disclosure; and FIG. **7** is a schematic structural diagram of a support component according to the embodiments of the present disclosure. In combination with FIG. **6** and FIG. **7**, the support component **1014** may include a strip-shaped support bar **1014***a***1**. One end a**11** of the support bar **1014***a***1** may be bonded to the support beam **10131**, and another end a**12** of the support bar **1014***a***1** may be fixedly connected to the bending portion **1021**.
(40) Optionally, FIG. **6** shows three support components **1014**. Certainly, the support device **101** according to the embodiments of the present disclosure may include more support components **1014**, and may for example include four or five support components **1014**. The number of the support components **1014** in the support device **101** is not limited in the embodiments of the present disclosure. The greater the number of the support components **1014** in the support device **101** is, the higher the reliability of supporting the bending portion **1021** of the bendable structure **102** by the support device **101** is.
(41) Optionally, another end a**12** of the support bar **1014***a***1** may be configured to be bonded to the bending portion **1021**, or configured to be clamped with the bending portion **1021**, which is not limited in the embodiments of the present disclosure as long as reliability of the connection between another end a**12** of the support bar **1014***a***1** and the bending portion **1021** can be ensured.
(42) FIG. **8** is a schematic structural diagram of another rotating shaft support according to the embodiments of the present disclosure. With reference to FIG. **8**, the support beam **10131** may be provided with at least one first groove **10131***b*. One end a**11** of each support bar **1014***a***1** may be disposed in one first groove **10131***b* and be bonded to first groove **10131***b*. By forming the first groove **10131***b* in the support beam **10131**, the support bar **1014***a***1** is bonded to the bottom surface of the first groove **10131***b*, such that the support bar **1014***a***1** can be prevented from shaking relative to the support beam **10131**, thereby ensuring the reliability of the connection between the support bar **1014***a***1** and the support beam **10131**.
(43) Optionally, the depth of the first groove **10131***b* may range from 0.4 mm to 0.8 mm, and the length of the support bar **1014***a***1** may range from 1.2 mm to 1.8 mm.
(44) Optionally, the number of the first grooves **10131***b* in the support beam **10131** may be equal to

the number of the support components **1014**. For example, three support components **1014** are shown in FIG. **6**, and correspondingly three first grooves **10131***b* are shown in FIG. **8**. Alternatively, the number of the first grooves **10131***b* in the support beam **10131** may also be greater than the number of the support components **1014**. Under this case, at least one of the first grooves **10131***b* in the support beam **10131** is not provided with the support bar **1014***a***1**.
(45) In the embodiments of the present disclosure, the orthographic projection of the end a**11** of the support bar **1014***a***1** on the support beam **10131** may be within the orthographic projection of another end a**12** of the support bar **1014***a***1** on the support beam **10131**. Referring to FIG. **7**, the length d**1** of the end a**11** of the support bar **1014***a***1** in the extending direction M of the support bar **1014***a***1** is equal to the length d**2** of another end a**12** of the support bar **1014***a***1** in the extending direction M of the support bar **1014***a***1**. The length d**3** of the end a**11** of the support bar **1014***a***1** perpendicular to the extending direction M of the support bar **1014***a***1** may be less than the length d**4** of another end a**12** of the support bar **1014***a***1** perpendicular to the extending direction M of the support bar **1014***a***1**. In this way, the orthographic projection of another end a**12** of the support bar **1014***a***1** on the support beam **10131** may have a bigger area than the orthographic projection of the end a**11** of the support bar **1014***a***1** on the support beam **10131**.
(46) Optionally, the length of the first groove **10131***b* perpendicular to the extending direction X of the support beam **10131** (i.e., the width of the first groove **10131***b*) may be less than the length of the support beam **10131** perpendicular to the extending direction X of the support beam **10131** (i.e., the width of the support beam **10131**). Moreover, in order that the end a**11** of the support bar **1014***a***1** is disposed in the first groove **10131***b*, the length d**3** of the end a**11** of the support bar **1014***a***1** needs to be less than or equal to the width of the first groove **10131***b*. Since the support beam **10131** generally has a small width, the first groove **10131***b* in the support beam **10131** also has a small width, such that the length d**3** of the end a**11** of the support bar **1014***a***1** also needs to be small.
(47) Additionally, since the length d**4** of another end a**12** of the support bar **1014***a***1** perpendicular to the extending direction M of the support bar **1014***a***1** is relatively large, the contact area between another end a**12** of the support bar **1014***a***1** and the bendable structure **102** can be increased, which can ensure the reliability of the connection between the support bar **1014***a***1** and the bendable structure **102**.
(48) Optionally, referring to FIG. **7**, the cross section of the support bar **1014***a***1** may be T-shaped. The cross section may be perpendicular to the extending direction M of the support bar **1014***a***1**. The smaller-sized end of the T-shape (i.e., the end a**11** of the support bar **1014***a***1**) may be bonded to the support beam **10131**, and the larger-sized end (i.e., another end a**12** of the support bar **1014***a***1**) may be configured to be fixedly connected to the bendable structure **102**.
(49) As another optional implementation, FIG. **9** is a schematic structural diagram of another support beam and support component according to the embodiments of the present disclosure; and FIG. **10** is a schematic structural diagram of another support component according to the embodiments of the present disclosure. In combination with FIG. **9** and FIG. **10**, each support component **1014** may include: a strip-shaped support bar **1014***b***1**, and a plurality of connectors **1014***b***2**. Each of the connectors **1014***b***2** may be bent-shaped. One end of each connector **1014***b***2** may be fixedly connected to the support bar **1014***b***1**, and another end may be connected to the support beam **10131**. For example, one end of the connector **1014***b***2** may be bonded to the support bar **1014***b***1**, and another end of the connector **1014***b***2** may be bonded to the support beam **10131**.
(50) In the embodiments of the present disclosure, at least two of the plurality of connectors **1014***b***2** may be disposed on different sides of the support bar **1014***b***1**. Since at least two of the connectors **1014***b***2** are disposed on different sides of the support bar **1014***b***1**, the at least two connectors **1014***b***2** may respectively connect different sides of the support bar **1014***b***1** to the support beam **10131**, thereby ensuring the stability of the connection between the support component **1014** and the support beam **10131**.

(51) Optionally, FIG. **9** shows three support components **1014**. Certainly, the support device **101** according to the embodiments of the present disclosure may also include more support components **1014**. For example, the support device **101** may include four or five support components **1014**. The number of the support components **1014** in the support device **101** is not limited in the embodiments of the present disclosure. The greater the number of the support components **1014** in the support device **101** is, the higher the reliability of supporting the bending portion **1021** of the bendable structure **102** by the support device **101** is.
(52) Additionally, the support component **1014** shown in FIG. **10** includes six connectors **1014***b***2**. Three connectors **1014***b***2** of the six connectors **1014***b***2** may be disposed on one side of the support bar **1014***b***1**, and the other three connectors **1014***b***2** may be disposed on the other side of the support bar **1014***b***1**. Certainly, the support component **1014** may also include other numbers of connectors **1014***b***2**, as long as it's ensured that at least two of the connectors **1014***b***2** are disposed on different sides of the support bar **1014***b***1**.
(53) FIG. **11** is a schematic structural diagram of still another rotating shaft support according to the embodiments of the present disclosure; and FIG. **12** is a schematic diagram of a partial structure of the rotating shaft support shown in FIG. **11**. Referring to FIGS. **11** and **12**, the support beam **10131** may be provided with a plurality of second grooves **10131***c*. Referring to FIG. **10**, each support component **1014** further includes a first protruding structure **1014***b***3** connected to another end of the connector **1014***b***2**. The first protruding structure **1014***b***3** may be disposed within one second grooves **10131***c*. That is, the second groove **10131***c* is formed in the support beam **10131** so as to connect the support beam **10131** and the support component **1014**.
(54) Optionally, the length w**1** of the first protruding structure **1014***b***3** perpendicular to the extending direction Z of the support bar **1014***b***1** ranges from 0.1 mm to 0.5 mm. Moreover, the orthographic projection of the second groove **10131***c* on the support beam **1013** may be greater than or equal to the size of the first protruding structure **1014***b***3**. For example, the difference between the length of each side of the orthographic projection of the second groove **10131***c* on the support beam **1013** and the length of the first protruding structure **1014***b***3** is 0.1 mm.
(55) Referring to FIG. **11**, nine second grooves **10131***c* may be formed on one side of the support beam **10131**, and nine second grooves **10131***c* may also be formed on the other side of the support beam **10131**. The number of the second grooves **1013***c* in the support beam **10131** may be equal to the number of the connectors **1014***b***2** of the support component **1014**. Certainly, the number of the second grooves **1013***e* in the support beam **10131** may also be greater than the number of the connectors **1014***b***2** of the support component **1014**, which is not limited in the embodiments of the present disclosure.
(56) Optionally, the support bar **1014***b***1**, the plurality of connectors **1014***b***2**, and the first protruding structures **1014***b***3** may be of an integral structure. For example, the support bar **1014***b***1**, the plurality of connectors **1014***b***2**, and the first protruding structures **1014***b***3** may be manufactured by the same manufacturing process.
(57) In the embodiments of the present disclosure, the support bar **1014***b***1** is further configured to be fixedly connected to the bending portion **1021** of the bendable structure **102**. For example, the support bar **1014***b***1** may be configured to be bonded to the bending portion **1021**, or may be configured to be clamped with the bending portion **1021**. The way of connection method between the support bar **1014***b***1** and the bending portion **1021** is not limited in the embodiments of the present disclosure as long as the reliability of the connection between the support bar **1014***b***1** and the bending portion **1021** can be ensured.
(58) In the embodiments of the present disclosure, the first protruding structure **1014***b***3** of the support component **1014** may be clamped into the second groove **10131***c* in the support beam **10131**, so as to achieve the fixed connection between the support component **1014** and the support beam **10131**. In addition, since the length w**1** of the first protruding structure **1014***b***3** is small, the support component **1014** may be directly pulled out of the second groove **10131***c* when the support

component **1014** is removed from the support beam **10131**.

(59) As another optional implementation, FIG. **13** is a schematic structural diagram of still another support beam and support component according to the embodiments of the present disclosure, and FIG. **14** is a schematic structural diagram of still another support component according to the embodiments of the present disclosure. In conjunction with FIGS. **13** and **14**, each support component **1014** may include: a strip-shaped support bar **1014***c***1**, a sheet structure **1014***c***2** connected to a side of the support bar **1014***c***1**, and a second protruding structure **1014***c***3** disposed at the end, distal from the support bar **1014***c***1**, of the sheet structure **1014***c***2**.

(60) Referring to FIG. **14**, the surface of the sheet structure **1014***c***2** which is not provided with the second protruding structure **1014***c***3** may be coplanar with one surface of the support bar. Certainly, the surface of the sheet structure **1014***c***2** which is not provided with the second protruding structure **1014***c***3** may also be not coplanar with one surface of the support bar **1014***c***1**, which is not limited in the embodiments of the present disclosure.

(61) FIG. **15** is a schematic structural diagram of yet another rotating shaft support according to the embodiments of the present disclosure, FIG. **16** is a schematic diagram of a partial structure of the support beam in the rotating shaft support shown in FIG. **15**, and FIG. **17** is a sectional view of the support beam in the rotating shaft support shown in FIG. **15** along the AA direction. In conjunction with FIGS. **15** to **17**, the support beam **10131** may be provided with a third groove **10131***d*, a fourth groove **10131***e*, and a communication groove **10131***f* for communicating the third groove **10131***d* and the fourth groove **10131***e* inside the support beam **10131**.

(62) FIG. **18** is a sectional view of the support beam and support component shown in FIG. **13** along the BB direction. Referring to FIG. **18**, the support bar **1014***c***1** may be provided in the third groove **10131***d*, the second protruding structure **1014***c***3** may be provided in the fourth groove **10131***e*, and the sheet structure **1014***c***2** may be provided in the communication groove **10131***f*, thereby connecting the support component **1014** to the support beam **10131**.

(63) The support bar **1014***c***1** may also be configured to be fixedly connected to the bending portion **1021** of the bendable structure **102**. For example, the support bar **1014***c***1** may be configured to be bonded to the bending portion **1021**, or may be configured to be clamped with the bending portion **1021**. The way of connection between the support bar **1014***c***1** and the bending portion **1021** is not limited in the embodiments of the present disclosure as long as the reliability of the connection between the support bar **1014***c***1** and the bending portion **1021** can be ensured.

(64) In the embodiments of the present disclosure, the support component **1014** may be mounted laterally on the support beam **10131**. The support component **1014** is placed in the third groove **10131***d* of the support beam **10131** first, and then a force is applied laterally to insert the second protruding structure **1014***c***3** of the support component **1014** into the fourth groove **10131***e* via the communication groove **10131***f*. When the support component **1014** is removed from the support beam **10131**, the second protruding structure **1014***c***3** may be pushed out of the fourth groove **10131***e* first, and then the support component **1014** is taken out of the third groove **10131***d*.

(65) The support component **1014** needs to be placed into or taken out of the third groove **10131***d* when the support component **1014** is mounted on or removed from the support beam **10131**. Therefore, the length of the third groove **10131***d* along the extending direction X of the support beam **10131** needs to be greater than the total length of the support bar **1014***c***1** and sheet structure **1014***c***2** of the support component **1014** along the extending direction N of the support bar **1014***c***1**.

(66) In the embodiments of the present disclosure, the support beam **10131** is provided with a rotating shaft hole **10131***a*. Thus, the bottom surface of the third groove **10131***d* and the bottom surface of the fourth groove **10131***e* are both arc surfaces, so as to prevent the rotating shaft in the rotating shaft hole **10131***a* from interacting with the support component **1014** in the third groove **10131***d*, the fourth groove **10131***e* and the communication groove **10131***f*, and ensure the reliability of the support device **101**. Therefore, in order to connect the support component **1014** to the support beam **10131**, both the surface of the support bar **1014***c***1** proximal to the support beam

10131 and the surface of the sheet structure **1014*c*2** proximal to the support beam **10131** need to be arc surfaces to match the shape of the third groove **10131*d*** and the shape of the fourth groove **10131*e***.
(67) The arc surface may be protruded towards the direction proximal to the bendable structure **102**, and the radius of the arc surface may be greater than the radius of the rotating shaft hole **10131*a***, so as to prevent the third groove **10131*d*** and the fourth groove **10131*e*** from communicating with the rotating shaft hole **10131*a***.
(68) Referring to FIGS. **5**, **6**, **8**, **9**, **11**, **13**, and **15**, the rotating shaft support **1013** may further include a strip-shaped support structure **10132** connected to the support beam **10131**, and two plate-shaped connecting structures **10133**. The two connecting structures **10133** may be connected to two ends of the support structure **10132**, respectively.
(69) The support structure **10132** provided in the rotating shaft support **1013** can support the support beam **10131**, to improve the rigidity of the rotating shaft support **1013** and avoid damages to the rotating shaft support **1013**, thereby ensuring the reliability of supporting the bending portion **1021** of the bendable structure **102** by the rotating shaft support **1013**.
(70) Referring to FIGS. **5**, **6**, **8**, **9**, **11**, **13**, and **15**, each connecting structure **10133** may at least be provided with a first through hole **10133*a***, and a second through hole **10133*b*** and a third through hole **10133*c*** which are disposed on two sides of the first through hole **10133*a***, respectively. The axis of the first through hole **10133*a*** may be co-linear with the axis of the rotating shaft hole **10131*a*** in the support beam **10131**, and the first through hole **10133*a*** may be connected to the rotating shaft. The second through hole **10133*b*** may be connected to the first support **1011**, and the third through hole **10133*e*** may be connected to the second support **1012**.
(71) In the case that the rotating shaft is mounted to the support beam **10131**, the rotating shaft may sequentially pass through the first through hole **10133*a*** in one of the two connecting structures **10133**, the rotating shaft hole **10131*a***, and the first through hole **10133*a*** of the other one of the two connecting structures **10133**. Moreover, the rotating shaft support **1013** is connected to the first support **1011** through the second through hole **10133*b*** in the connecting structure **10133** of the rotating shaft support **1013**, such that the rotating shaft support **1013** rotates relative to the first support **1011**. The rotating shaft support **1013** is connected to the second support **1012** through the third through hole **10133*c*** in the connecting structure **10133** of the rotating shaft support **1013**, such that the rotating shaft support **1013** rotates relative to the second support **1012**.
(72) FIG. **19** is a schematic structural diagram of another support device according to the embodiments of the present disclosure. Referring to FIG. **19**, the support device **101** may further include a first bracket **1015** and a second bracket **1016**. One end of the first bracket **1015** may be rotatably connected to the rotating shaft support **1013**, and another end of the first bracket **1015** may be fixedly connected to the first support **1011**. One end of the second bracket **1016** may be rotatably connected to the rotating shaft support **1013**, and another end of the second bracket **1016** may be fixedly connected to the second support **1012**. That is, the rotating shaft support **1013** may be rotatably connected to the first support **1011** through the first bracket **1015** and rotatably connected to the second support **1012** through the second bracket **1016**.
(73) Optionally, the first bracket **1015** may be rotatably connected to the rotating shaft support **1014** by gears or hinges, and the second bracket **1016** may be rotatably connected to the rotating shaft support **1014** by gears or hinges. The first bracket **1015** and the second bracket **1016** may be connected to the rotating shaft support **1014** in the same manner or in different manners, which is not limited in the embodiments of the present disclosure.
(74) Referring to FIG. **19**, it can be seen that the first support **1011** and the second support **1012** may both be plate-shaped structures, and the first support **1011** may be provided with a first slot **1011*a*** and the second support **1012** may be provided with a second slot **1012*a***. The bendable structure **102** may be disposed in the first slot **1011*a*** and the second slot **1012*a***, and the first slot **1011*a*** and the second slot **1012*a*** may be configured to protect the sides of the bendable structure

102 to avoid damages to the sides of the bendable structure **102**.
(75) In summary, the embodiments of the present application provide a support device. The rotating shaft support and each support component in the support device can provide support for the bending portion of the bendable structure. Moreover, the first support in the support device can provide support for the first non-bending portion of the bendable structure, and the second support in the support device can provide support for the second non-bending portion of the bendable structure. That is, the support device according to the embodiments of the present disclosure can be configured to support the entire bendable structure, which can prevent the bending portion of the bendable structure from bulging, thereby ensuring the flatness of the bendable structure.
(76) FIG. **20** is a schematic structural diagram of a foldable device according to the embodiments of the present disclosure, and FIG. **21** is an exploded schematic diagram of the foldable device shown in FIG. **20**. Referring to FIGS. **20** and **21**, the foldable device **10** may include: a bendable structure **102** and the support device **101** as provided in the aforesaid embodiments. The support device **101** may be disposed on the back side of the bendable structure **102**, and the support device **101** is connected to the back side of the bendable structure **102**.
(77) The support device **101** may be configured to support the entire bendable structure **102**, which can prevent the bending portion **1021** of the bendable structure **102** from bulging, thereby ensuring the flatness of the bendable structure **102**. In addition, since the support component **1014** in the support device **101** is connected to the bending portion **1021** of the bendable structure **102**, the bendable structure **102** can be prevented from sliding relative to other devices in the bending process of the foldable device **10**, thereby ensuring the stability of the foldable device **10**.
(78) Optionally, the bendable structure **102** may be a flexible display panel. Accordingly, the foldable device **10** may be a foldable display device. Certainly, the bendable structure **102** may also be other bendable structures, which is not limited in the embodiments of the present disclosure.
(79) Referring to FIG. **21**, the foldable device **10** may further include a reinforcement plate **103**. The support device **101** and the bendable structure **102** may be connected by the reinforcement plate **103**.
(80) Optionally, the reinforcement plate **103** may include: a bending region **103***a* and a first non-bending region **103***b* and a second non-bending region **103***c* disposed on two sides of the bending region **103***a*, respectively. The bending region **103***a* may be connected to the bending portion **102** of the bendable structure **102**, the first non-bending region **103***b* may be connected to the first non-bending portion **1022** of the bendable structure **102**, and the second non-bending region **103***c* may be connected to the second non-bending portion **1023** of the bendable structure **102**. Furthermore, the bending region **103***a* of the reinforcement plate **103** is further bonded to the support bar in the support device **101**.
(81) In the embodiments of the present disclosure, the reinforcement plate **103** may be made of a metallic material so as to provide reinforcement to the bendable structure **102**, thereby improving the strength of the bendable structure **102**. Moreover, since the bending region **103***a* of the reinforcement plate **103** needs to be bent, the bend region **103***a* may be provided with a plurality of openings (i.e., the bending region may be hollowed out) to reduce the stress in the bending region **103***a*. In addition, the first non-bending region **103***b* and the second non-bending region **103***c* may be sheet-shaped.
(82) Optionally, referring to FIG. **21**, the openings in the bending region **103** of the reinforcement plate **103** may be round. Certainly, the openings in the bending region **103** of the reinforcement plate **103** may also be in other shapes, and may for example be rectangular, which is not limited in the embodiments of the present disclosure.
(83) Referring to FIG. **21**, the foldable device **10** may further include: a first protective structure **104**, a second protective structure **105**, and a third protective structure **106**. The first protective structure **104** may be disposed on the side, distal from the bendable structure **102**, of the rotating shaft support **1013** of the support device **101**, and is configured to wrap the rotating shaft support

1013 to avoid damages to the rotating shaft support **1013**. The second protective structure **105** may be disposed on the side, distal from the bendable structure **102**, of the first support **1011** of the support device **101**, and is configured to wrap the first support **1011** to avoid damages to the first support **1011**. The third protective structure **106** may be disposed on the side, distal from the bendable structure **102**, of the second support **1012** of the support device **101**, and is configured to wrap the second support **1012** to avoid damages to the second support **1012**.
(84) Optionally, the foldable display device may be any product or component having a display function, such as electronic paper, an organic light-emitting diode (OLED) display device, a mobile phone, a tablet computer, a TV, a display, a notebook computer, a digital photo frame, a navigator, and the like.
(85) In summary, the embodiments of the present disclosure provide a foldable device. The foldable device includes a bendable structure and a support device disposed on the back side of the bendable structure. The rotating shaft support and each support component in the support device can support the bending portion of the bendable structure. Moreover, the first support in the support device can support the first non-bending portion of the bendable structure, and the second support can support the second non-bending portion of the bendable structure. That is, the support device can be configured to support the entire bendable structure, which can prevent the bending portion of the bendable structure from bulging, thereby ensuring the flatness of the bendable structure.
(86) The descriptions above are merely optional embodiments of the present disclosure, and are not intended to limit the present disclosure. Within the spirit and principles of the disclosure, any modifications, equivalent substitutions, improvements, and the like are within the protection scope of the present disclosure.

Claims

1. A support device, wherein the support device is disposed on a back side of a bendable structure, the bendable structure comprising a bending portion, and a first non-bending portion and a second non-bending portion disposed on two sides of the bending portion, respectively; the support device comprising: a first support, configured to be connected to the first non-bending portion; a second support, configured to be connected to the second non-bending portion; a rotating shaft support, configured to be rotatably connected to the first support and the second support, wherein the rotating shaft support comprises a strip-shaped support beam; and at least two support components, wherein the at least two support components are arranged at intervals on the support beam, and each of the at least two support components comprises a strip-shaped support bar, wherein the support bar is connected to the support beam and configured to be fixedly connected to the bending portion.
2. The support device according to claim 1, wherein the rotating shaft support comprises a strip-shaped rotating shaft; wherein the support beam is provided with a rotating shaft hole penetrating through the support beam, wherein an axis of the rotating shaft hole is parallel to an extending direction of the support beam; and the rotating shaft is disposed in the rotating shaft hole and is rotatable in the rotating shaft hole.
3. The support device according to claim 2, wherein one end of the support bar is bonded to the support beam, and another end of the support bar is configured to be fixedly connected to the bending portion.
4. The support device according to claim 3, wherein the support beam is provided with at least one first groove, wherein one end of each support bar is disposed in one first groove and is bonded to the first groove.
5. The support device according to claim 3, wherein an orthographic projection of one end of the support bar on the support beam is within an orthographic projection of another end of the support bar on the support beam.

6. The support device according to claim 5, wherein the support bar has a T-shaped cross section, and the cross section is perpendicular to an extending direction of the support bar.
7. The support device according to claim 2, wherein each of the at least two support components further comprises a plurality of connectors, each of the connectors being curved; wherein one end of each of the connectors is fixedly connected to the support bar, and another end of the connector is connected to the support beam, and at least two connectors of the plurality of connectors are disposed on different sides of the support bar.
8. The support device according to claim 7, wherein the support beam is provided with a plurality of second grooves, and each of the at least two support components further comprises a first protruding structure connected to another end of the connector, the first protruding structure being disposed in one of the second grooves.
9. The support device according to claim 2, wherein each of the at least two support components further comprises a sheet structure connected to a side of the support bar and a second protruding structure disposed at an end, distal from the support bar, of the sheet structure; the support beam is provided with a third groove, a fourth groove, and a communication groove disposed inside the support beam and configured to communicate the third groove and the fourth groove; and wherein the support bar is disposed in the third groove, the second protruding structure is disposed in the fourth groove, and the sheet structure is disposed in the communication groove.
10. The support device according to claim 9, wherein a length of the third groove along the extending direction of the support beam is greater than a total length of the support bar and the sheet structure along an extending direction of the support bar.
11. The support device according to claim 9, wherein a bottom surface of the third groove and a bottom surface of the fourth groove are both arc surfaces, and a surface, proximal to the support beam, of the support bar and a surface, proximal to the support beam, of the sheet structure are both the arc surfaces; wherein the arc surfaces are protruded in a direction proximal to the bendable structure, and a radius of the arc surfaces is greater than a radius of the rotating shaft hole.
12. The support device according to claim 2, wherein the rotating shaft support further comprises: a strip-shaped support structure connected to the support beam and two plate-shaped connecting structures, the two connecting structures being connected to two ends of the support structure, respectively; wherein each of the connecting structures is at least provided with a first through hole, and a second through hole and a third through hole disposed on two sides of the first through hole respectively, wherein an axis of the first through hole is co-linear with an axis of the rotating shaft hole, the first through hole is connected to the rotating shaft, the second through hole is connected to the first support, and the third through hole is connected to the second support.
13. The support device according to claim 1, further comprising: a first bracket and a second bracket; wherein one end of the first bracket is rotatably connected to the rotating shaft support, and another end of the first bracket is fixedly connected to the first support; and one end of the second bracket is rotatably connected to the rotating shaft support, and another end of the second bracket is fixedly connected to the second support.
14. A foldable device, comprising: a bendable structure and at least one support device; wherein the support device is disposed on and connected to a back side of the bendable structure, the bendable structure comprising a bending portion, and a first non-bending portion and a second non-bending portion disposed on two sides of the bending portion, respectively; the support device comprising: a first support, configured to be connected to the first non-bending portion; a second support, configured to be connected to the second non-bending portion; a rotating shaft support, configured to be rotatably connected to the first support and the second support, wherein the rotating shaft support comprises a strip-shaped support beam; and at least two support components, wherein the at least two support components are arranged at intervals on the support beam, and each of the at least two support components comprises a strip-shaped support bar, wherein the support bar is connected to the support beam and configured to be fixedly connected to the bending portion.

15. The foldable device according to claim 14, wherein the bendable structure is a flexible display panel.
16. The foldable device according to claim 14, further comprising: a reinforcement plate; wherein the support device and the bendable structure are connected by the reinforcement plate.
17. The foldable device according to claim 14, wherein the rotating shaft support comprises a strip-shaped rotating shaft; wherein the support beam is provided with a rotating shaft hole penetrating through the support beam, wherein an axis of the rotating shaft hole is parallel to an extending direction of the support beam; and the rotating shaft is disposed in the rotating shaft hole and is rotatable in the rotating shaft hole.
18. The foldable device according to claim 17, wherein one end of the support bar is bonded to the support beam, and another end of the support bar is configured to be fixedly connected to the bending portion.
19. The foldable device according to claim 18, wherein the support beam is provided with at least one first groove, wherein one end of each support bar is disposed in one first groove and is bonded to the first groove.
20. The foldable device according to claim 18, wherein an orthographic projection of one end of the support bar on the support beam is within an orthographic projection of another end of the support bar on the support beam.